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Inventor(s)	Gelin; Jean-Luc et al.

Taste-modifying ingredient

Abstract

Use of 7-[4,5-dihydroxy-6-(hydroxymethyl)-3-(3,4,5-trihydroxy-6-methyloxan-2-yl)oxyoxan-2-yl]oxy-5-hydroxy-2-(4-hydroxyphenyl)chromen-4-one (rhoifolin) as a taste modifying ingredient in order to reduce bitterness and/or increase sweetness of a foodstuff or beverage.

Inventors: **Gelin; Jean-Luc (Satigny, CH), Delattre; Maxime (Satigny, CH)**

Applicant: **Firmenich SA** (Satigny, CH)

Family ID: **1000008780313**

Assignee: **Firmenich SA (Satigny, CH)**

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Primary Examiner: Mornhinweg; Jeffrey P

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a continuation of U.S. patent application Ser. No. 16/165,165, filed Oct. 19, 2018, which is a continuation of U.S. patent application Ser. No. 15/967,905, filed May 1, 2018, which is a continuation of U.S. patent application Ser. No. 14/901,389, filed Dec. 28, 2015, which is a United States national stage application of PCT Application No. PCT/EP2014/061417, filed Jun. 3, 2014, which claims the benefit of priority of U.S. Provisional Application No. 61/840,401, filed Jun. 27, 2013.

TECHNICAL FIELD
(1) The present invention relates to a taste-modifying ingredient. The invention further relates to the use of the taste-modifying ingredients in a foodstuff and a taste-modifying composition comprising the taste-modifying ingredient.
BACKGROUND AND PRIOR ART
(2) Sugar is a popular sweetening additive in human food preparation. By sugar is understood

sucrose but also other commonly used calorie rich sweetening additives such as glucose, fructose and high fructose corn syrups. Popular feeding habits tend to show an over consumption of sugar even though it is well established that this is a known cause of various adverse effects on health, the most common including tooth decay and obesity.
(3) To date, various products have been proposed which seek to address these problems. For instance, artificial high intensity sweeteners have been developed which deliver a sweet taste at very low doses. Of the high intensity sweeteners already present on the market, Sucralose®, Aspartame, Potassium Acesulfame, cyclamate, saccharine can be named as well known alternatives. However, there is a strong desire by an ever-increasing number of consumers for natural or naturally derived products in preference to their artificial counterparts. Thus, it would be highly desirable to provide a product which meets this consumer need.
(4) Within the class of naturally occurring sweeteners, a growing number of products is becoming available. Examples include thaumatin, luo han guo, brazzein, curculin, glycyrrhizin and stevia.
(5) Rhoifolin is a compound found in the juice of citrus bergamia (bergamot). To the best of our knowledge, this compound has never been identified as a taste-modifying ingredient for increasing sweetness.
(6) It would also be desirable to reduce the bitterness of certain foodstuffs and beverages. To the best of our knowledge, rhoifolin has never been identified as a taste-modifying ingredient for reducing bitterness.
SUMMARY OF THE INVENTION
(7) Thus, according to the present invention, there is provided the use of 7-[4,5-dihydroxy-6-(hydroxymethyl)-3-(3,4,5-trihydroxy-6-methyloxan-2-yl)oxyoxan-2-yl]oxy-5-hydroxy-2-(4-hydroxyphenyl)chromen-4-one as a taste modifying ingredient.
(8) For the purposes of the present invention, 7-[4,5-dihydroxy-6-(hydroxymethyl)-3-(3,4,5-trihydroxy-6-methyloxan-2-yl)oxyoxan-2-yl]oxy-5-hydroxy-2-(4-hydroxyphenyl)chromen-4-one is also referred to herein as “rhoifolin”.
(9) The invention further provides a taste modifying composition comprising an effective amount of rhoifolin.
(10) The invention also provides a method of reducing or masking bitterness of a foodstuff or beverage by adding rhoifolin thereto.
(11) The invention also provides a method of increasing the sweetness of a foodstuff or beverage by adding rhoifolin thereto.

Description

DETAILED DESCRIPTION OF THE INVENTION
(1) In a foodstuff or beverage, rhoifolin may be present in amounts within the ranges of from 5 to 300 ppm.
(2) Preferably rhoifolin is present at a concentration of from 5 to 200 ppm, more preferably 5 to 150 ppm, most preferably from 10 to 100 ppm.
(3) Rhoifolin is commercially available and can be purchased from, for example, Interquim, Spain.
(4) Rhoifolin can be used in a many products where there is a desire to reduce bitterness and/or to increase sweetness. Examples include but are not limited to tea, coffee, fruit juice and fruit-flavoured beverages, jams and jellies, peanut butter, pies, puddings, cereals, candies, ice creams, yogurts, bakery products; health care products, such as toothpastes, mouthwashes, cough drops, cough syrups; chewing gums; and sugar substitutes.
(5) The invention will now be illustrated with reference to the following examples. All amounts are % by weight unless otherwise indicated.
EXAMPLES

Example 1

(6) Application of Rhoifolin in Grapefruit Juice

(7) Various samples were prepared using the ingredients shown in the table below (the amounts are % by weight).

(8) TABLE-US-00001 TABLE 1 Sample Grapefruit Juice (1) Rhoifolin (2) 5 99.9 0.1 6 99.6 0.4 7 99.2 0.8 8 98.8 1.2 9 98.4 1.6 10 98.0 2.0 B 100 0 (1) Eco+, purchased at Leclerc, France (2) ex Interquim, Spain

(9) The rhoifolin was weighed into sufficient propylene glycol (PG) to allow for full dissolution and heated in a water bath at 40° C. for 30 minutes until the rhoifolin was fully solubilised. The grapefruit juice was then mixed with the rhoifolin solution and the mixtures were kept at room temperature for 1 hour before sensory tests.

(10) The sensory tests were performed as follows: 6 trained panellists were asked to rate the sweetness, bitterness and flavor intensity of the samples prepared above under blind conditions using a 5 point intensity rating scale where 0 indicates no intensity and 5 indicates extremely intense.

(11) Average intensity scores were treated for mean comparison Duncan testing using a FIZZ statistic software.

(12) The results are given in the following table:

(13) TABLE-US-00002 TABLE 2 Bitterness Sweetness Flavor Rhoifolin Sample Intensity Intensity Intensity (ppm) B 4.17 2.25 3.50 0 5 3.83 2.42 3.67 5 6 3.33 2.50 3.33 20 7 3.08 2.75 3.25 40 8 3.08 3.17 3.08 60 9 2.67 3.50 3.17 80 10 2.33 4.00 3.33 100

(14) The results demonstrate that addition of rhoifolin reduces bitterness and increased sweetness whilst leaving the flavor intensity substantially unchanged.

Example 2

(15) Application of Rhoifolin in a Sucrose-Free Coffee Product

(16) Various samples were prepared using the ingredients shown in the table below (the amounts are % by weight).

(17) TABLE-US-00003 TABLE 3 Sample Water (1) Instant soluble Coffee (2) Rhoifolin (3) 1 99.4 0.5 0.1 2 99.55 0.05 0.4 3 98.7 0.5 0.8 4 98.3 0.5 1.2 A 100 0 0 (1) Still Mineral Water, ex Arkina, Switzerland (2) Commercially available instant soluble coffee, purchased in Migros, Switzerland (3) ex Interquim, Spain

(18) The rhoifolin was weighed into sufficient propylene glycol (PG) to allow for full dissolution and heated in a water bath at 40° C. for 30 minutes until the rhoifolin was fully solubilised.

(19) The instant soluble coffee was weighed and incorporated into the water. Solubilisation was achieved at room temperature.

(20) The mixtures were kept at room temperature for 1 hour before sensory tests.

(21) The sensory tests were performed as follows: 6 trained panellists were asked to rate the sweetness, bitterness and coffee-like intensity of the samples prepared above under blind conditions using a 5 point intensity rating scale where 0 indicates no intensity and 5 indicates extremely intense.

(22) Average intensity scores were treated for mean comparison Duncan testing using a FIZZ statistic software.

(23) The results are given in the following table:

(24) TABLE-US-00004 TABLE 4 Bitterness Sweetness Coffee-like Rhoifolin Sample Intensity Intensity Intensity (ppm) A 3.33 0.33 2.83 0 1 3.33 0.33 2.83 5 2 2.58 0.67 2.92 20 3 2.33 1.83 2.75 40 4 2.00 2.58 2.33 60

(25) The results demonstrate that it is preferable that more than 5 ppm of rhoifolin is present for certain applications.

Claims

1. A method of reducing bitterness of a foodstuff or beverage, the method comprising introducing a taste-modifying ingredient to a foodstuff or beverage in an amount effective to reduce a bitter taste, wherein the taste-modifying ingredient is 7-[4,5-dihydroxy-6-(hydroxymethyl)-3-(3,4,5-trihydroxy-6-methyloxan-2-yl)oxyoxan-2-yl]-oxy-5-hydroxy-2-(4-hydroxyphenyl)-chromen-4-one.
2. The method of claim 1, wherein the taste modifying ingredient is present in the foodstuff or beverage at a concentration ranging from 5 to 100 ppm.
3. A method of enhancing sweetness of a foodstuff or beverage, the method comprising introducing a taste-modifying ingredient to a foodstuff or beverage in an amount effective to enhance a sweet taste, wherein the taste-modifying ingredient is 7-[4,5-dihydroxy-6-(hydroxymethyl)-3-(3,4,5-trihydroxy-6-methyloxan-2-yl)oxyoxan-2-yl]-oxy-5-hydroxy-2-(4-hydroxyphenyl)-chromen-4-one.
4. The method of claim 3, wherein the taste modifying ingredient is present in the foodstuff or beverage at a concentration ranging from 5 to 100 ppm.